

The Effects of Online Interactive Games on High School Students' Achievement and Motivation in History Learning

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ABSTRACT

Many studies demonstrate that Digital Game Based Learning (DGBL) can foster learning effect. The purpose of this study is to survey whether the online game in junior high school students can encourage learning effect in Taiwan's History. So, the research applied Interactive Game-based Learning System (IGLS) to junior high history teaching as an ancillary tool to enhance users' learning motivation and academic performance. After a four-week experiment conducted on eleven students, the paired sample t test analysis considerably impact on experimental group to significance .05, and one-way ANCOVA showed login IGLS the number of days every week more learning effect to be better. Besides, questionnaires exhibited that IGLS not only magnetized over ninety percent of students' participation but invigorated their learning will. The research presents positive results on enhancing students' learning motivation and academic performance.

Keywords: Digital Game Based Learning, History Education, Learning Motivation, Online Game, Taiwan

INTRODUCTION

With rapid growth in information communication technology, online games have gradually become part of people's life in recent years. It is believed that digital games, as a catalyst, can be very useful teaching media. According to the Taiwan Institute for Information Industry in 2010 published report of using broadband net-

work. Taiwan's Internet users most commonly used online gaming services (50.28%), the largest proportion of 12-19 years accounting for 83.08%-90.91%. For this reason, teachers must elevate students' learning ability and knowledge through temporary assistance like information technology. To teachers and students, the meaning of this incoming digital trend counts for more challenges than opportunities. History, a subject about facts, should also be inextricably linked to digital world. Being a junior high history

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teacher, the researcher found that it would be better to use a lively way-learning by playing-we can use some digital assistance to arouse students' learning motivation and academic performance. Moreover, traditionally teacher-centered didactic instruction is progressively substituted by student-centered various interactive learning patterns. The innovating digital learning which claims for enhancing learners' motivation and initiative exploration is available to all. Terms like E-learning, U-learning, Digital Game-based Learning (DGBL) gradually represent information technology integrated into teaching in school. DGBL was first proposed by Marc Prensky in 2001. Through digital games, learners obtain sense of achievement in solving problems and dealing with challenges. Digital game based learning takes both entertainment and education into consideration. In Prensky's opinion, digital game based learning will be the main learning pattern in the 21st century, the traditional learning style such as emphasizing reading and writing is going to be replaced. Online games stand for part of teenagers' culture, so it's undoubtedly inevitable to integrate it into school education.

Moreover, successful learning definitely needs motivation. Learning motivation is the key factor whether digital game based learning and web base learning community work or not. Munn et al. (1969) defined that motivation, as a sort of intrinsic power such as interest, attitude and desire, can influence personal behavior. On the other hand, Keller (1983) regarded motivation as people's choice of experiences and goals, and how they make efforts to achieve them. Besides, Reilly (1983), Lewis, and Tanner considered motivation to be the power inside oneself. Of all digital learning motivation theory, Keller's (1983) ARCS theory is the most universal one. The theory is composed of four aspects: attention, relevance, confidence, and satisfaction. Taran (2005) placed specific emphasis on attention, so he promoted ten skills for maintaining learners' online learning attention to strengthen motivation. That proves the importance of motivation to academic performance. Paras and Bizzocchi (2005) combined

ARCS theory with Csikszentmihalyi's (1991) Flow theory because they thought that a good digital game design immerses learners in games, and then brings about effective learning process.

Games, one of the oldest learning methods, are part of human cognition development. They have positive effect on learning basic cognition operation, understanding complex grammar structure, and abstract thinking. According to Vygotsky (1966), learning by playing helped promote Zone of Proximal Development (ZPD). In Bruner's (1960, 1989) research, it indicates that games can adjust teaching for both fast learners and slow ones to attain the spirit of spiral curriculum. Cheng and Chen (2008) also found that peer cooperation and coordination through digital interactive games enriched learners' learning motivation and academic performance. Prensky (2003), a game designer, pointed out the integration of games and learning will change from the traditional idea "learning is depressing" to "learning by playing, learning through playing." To realize this ideal purpose, letting students learn actively will make the process of learning become more and more interesting.

After Lippitt (1981) introduced the term "learning community," most scholars and experts generally acknowledge that it is characterized by learning by sharing. Web base learning community, online virtual space, means a group of learners form common learning culture in which they brainstorm, make efforts and understand each other. By integrating scholars' viewpoints of online learning community, Wang (2003) offered a clear definition: web base learning community is a kind of computer-mediated communication which includes sharing ideas, knowledge, experience and information, forming common purpose, investigating, and enhancing knowledge and ability. Within the atmosphere of support, care, openness and compliance, learners can challenge his/her own belief, value, knowledge and experience. Furthermore, learners enhance his/ her knowledge and ability as a result of introspection. The learning culture figures that learning relies on social interaction which derives from personal active communication, sharing and investiga-

tion. In this study, looking through IGLS digital game-based learning platform for entertaining, so that groups of learners in the free state, the natural formation of learning communities, the game in the team members, through chat rooms to share and explore the history knowledge, create cooperative learning and knowledge sharing of consciousness.

Digital learning can enhance students' motivation except for focusing too much on playing games instead of learning itself. In Mitchell and Savill-Smith's (2004) study, if digital learning turns into games addiction, game type learning will bring down learners' motivation. So, Johnson and Johnson (1991) propound to take interest, collaboration, feedback and assessment into consideration. Prensky featured six key elements to choose or develop digital game based learning patterns: (1) rules (2) goals or objectives (3) outcomes and feedback (4) conflict, competition, challenge and opposition (5) interaction (6) representation and story. For teachers, these ideas are great references to establish or adopt digital game patterns. In short, teachers in the design or selection of digital game should be based on student-centered, attention to the goal of the game itself and interaction, and create a learning environment of healthy competition and the game in the process of assessment and immediate feedback capabilities.

To put these ideals into reality, an experiment of IGLS was conducted for a month, surveys toward participants and interviews were both applied. Employing IGLS as an ancillary tool in learning Taiwan history, the researcher looks forward to enhancing learners' motivation and academic performance by online community learning mechanism like group competition. The research enables learners to form community through free grouping, and members of the group probe into the knowledge of history and share information with each other in chat rooms. Additionally, they create the sense of collaboration as well as knowledge sharing.

IGLS FOR HISTORY EDUCATIONAL GAMES DESIGN

Interactive Game-Based Learning System, IGLS

The IGLS system is composed of three elements, which are quiz game module, item generation module, and learning management module. The system not only enables students to review their learning, but also to compete against other participants. To increase the randomness of quizzes, the system offers three types of item generation algorithms, including multiple choice with random options, match items, and random argument computational problems. In doing so, it is expected to effectively attract students' attention. Furthermore, the learning management modules allow students and teachers to monitor the learning process. This pedagogical experiment is a design for evaluating students' motivation and academic performance.

Game Design

IGLS's user interface includes manager (teacher) and player (learner). Players can register and, after checked by managers, choose their own courses. They can also pace themselves to choose stand-alone or online version. After signing in, players can select units and start a game as long as both sides have equal team members (2 vs. 2 or 3 vs. 3...etc.).

Both teams have one hundred hit points. After showing randomly selected questions, players attack any opponent or choose compound attack. If the opponent gets the answer wrong, its hit points will diminish. On the contrary, if it gets the answer right, the opponent's hit points reduce. Game is over when one loses all hit points.

Players review records in the form of a chart which shows each unit's score. Furthermore, they can evaluate themselves. Besides, they create the sense of collaboration as well

as knowledge sharing. Managers can check learners' performance by viewing each record, and then make adjustments.

Teaching Materials

The experimental group learned about Taiwan history within games. IGLS includes context of seven graders' sociology textbook "Taiwan history (Vol. 1), and the contents are prehistory, international competition, Cheng's governance and earlier stage of Ching dynasty's occupation.

EXPERIMENT DESIGN AND RESULT

Experiment Design

The study took sample of thirty students from a certain Taichung Municipal Junior High School. Based on the pretest average score 90, nineteen students who got higher scores are in the control group (high achievers), whereas another eleven students who got lower scores are in the experimental group (low achievers).

A pretest on Taiwan history (Vol. 1) was conducted a week before the experiment began. In the first week, I took students were taken to the computer classroom, introduced to the system and instructed the way to use it. When the experiment commenced, I reminded them to use it after school every week. The experiment lasted for a month, and students can use the system wherever Internet is accessible. In the fourth week, a survey was done to the experimental group. After the experiment ended (the fifth week), a post test was conducted to all students.

To find out whether there are remarkable improvements in academic performance or not, paired sample t test analysis of the pretest and post test of Taiwan history (Vol. 1) on both groups was conducted. Thus, the pretest result of Taiwan history (Vol. 1) is used as covariance, learning method as independent variable, the post test of Taiwan history (Vol. 1) as dependent variable, to determine if the factors indicate significant influence on academic performance. Consequently, by analyzing experimental group's survey, students' opinion and their intention to participate were identified.

Taiwan History (Vol. 1) Achievement Test

In order to evaluate the academic performance between experimental and control group, the research drew up test questions about Taiwan history (Vol. 1) which came from both IGLS's contents and Taiwan history textbook 1. The sample contains four hundred multiple-choice questions.

Paired Sample Comparison for Control Group and Experimental Group

Table 1 contains statistics of the experiment which is conducted by unequal grouping's pretest and post test. In order to examine the improvement between pretest and post test on both groups, paired sample t test is used. In Table 2, significant progression in each group can be discovered. Furthermore, the experimental group students made better improvement than the control group (-7.27 to -1.68). With assistance of IGLS, the experimental group

Table 1. Descriptive statistics for achievement

		Mean	N	SD	SEM
Control group	Pretest1	94.5263	19	3.04354	.69824
	Posttest1	96.2105	19	2.57291	.59027
Experimental group	Pretest2	84.1818	11	3.51620	1.06017
	Posttest2	91.4545	11	10.04354	3.02824

Table 2. Paired sample *t* test analysis for achievement

		Pooled Variance					t	DF	Significance (Two-Tailed)
		Mean	SD	SEM	95% CI				
					Low	High			
Control group	pretest-posttest	-1.68421	3.66746	.84137	-3.45187	.08345	-2.002	18	.061
Experimental group	pretest-posttest	-7.27273	7.55104	2.27672	-12.34558	-2.19987	-3.194	10	.010

students demonstrated level of significance at .05 ($P=.010$), while the control group students indicated insignificant progression ($P=.061$). As a result, the system indeed promotes students' academic performance in Taiwan history.

Affecting Factors of the Learning of Taiwan's History

The researchers measured factors in order to exclude the former experimental results, the history of Taiwan's pre-test achievement test scores as covariates. Respectively, the experimental group of gender, weekly time of surfing the Net, affection network leisure type and login IGLS per week as the independent variable. The post-test scores of Taiwan's history achievement test are as dependent variables.

Using ANCOVA analysis detects the impact of the factors in Taiwan's history learning. Exclude the impact of pre-test scores for Taiwan's history of learning is of significant impact.

According to the ANCOVA test of Tables 3 through 6, the statistical results showed that only the effect of the amount of days for student of logging in the IGLS system is with .041 (<0.05) significant statistical difference on the post-test scores. The effects of other independent variable on the post-test scores did not reach significant statistical difference. Statistical inference based on the results of this study, The learning outcome of Taiwan's history for the students of experimental group are influenced by the amount of days for logging in the IGLS system.

Table 3. Gender on learning effect of the ANCOVA analysis

Source	Type III Sum of Squares	Squares	df	Mean Square	F
Corrected Model	22.020(a)	2	11.010	2.380	.163
Intercept	47.500	1	47.500	10.269	.015
Experimental group pretest	.001	1	.001	.000	.989
gender	16.900	1	16.900	3.653	.098
Error	32.380	7	4.626		
Total	89168.000	10			
Corrected Total	54.400	9			

Dependent Variable: Experimental group post-test

a R Squared = .405 (Adjusted R Squared = .235), $p<.05$

Table 4. Weekly time of surfing the Net on learning effect of the ANCOVA analysis

Source	Type III Sum of Squares	Squares	df	Mean Square	F
Corrected Model	8.257(a)	3	2.752	.358	.786
Intercept	23.996	1	23.996	3.120	.128
Experimental group pretest	2.857	1	2.857	.372	.565
weekly time of surfing the Net	3.137	2	1.569	.204	.821
Error	46.143	6	7.690		
Total	89168.000	10			
Corrected Total	54.400	9			

Dependent Variable: Experimental group post-test
a R Squared = .152 (Adjusted R Squared = -.272) $p < .05$

Table 5. Affection network leisure type on learning effect of the ANCOVA analysis

Source	Type III Sum of Squares	Squares	df	Mean Square	F
Corrected Model	29.431(a)	3	9.810	2.357	.171
Intercept	37.614	1	37.614	9.039	.024
Experimental group pretest	.031	1	.031	.008	.934
Affection network leisure type	24.311	2	12.156	2.921	.130
Error	24.969	6	4.161		
Total	89168.000	10			
Corrected Total	54.400	9			

Dependent Variable: Experimental group post-test
a R Squared = .541 (Adjusted R Squared = .312) $p < .05$

IGLS Questionnaire

The research applies Likert's Five-Point scale to find out experimental group students' opinion and motivation of using IGLS. Strongly agree gets five points, agree gets four, neutral gets three, disagree gets two and strongly disagree gets one point. In accordance with the function and design, the researcher discussed and analyzed with professor and program designers to keep content validity.

To understand the function of IGLS and the motivation it triggers, the research surveyed the experimental group students in the fourth week. The result can be future's reference to the design of interface or the selection of the game platform. The research uses Likert's Five-Point scale: strongly agree gets five points, agree gets four, neutral gets three, disagree gets two and strongly disagree gets one point. And the whole survey is divided into three parts: interface, function and motivation.

Table 6. Login IGLS per week on learning effect of the ANCOVA analysis

Source	Type III Sum of Squares	Squares	df	Mean Square	F
Corrected Model	37.382(a)	3	12.461	4.393	.059
Intercept	32.463	1	32.463	11.445	.015
Experimental group pretest	.982	1	.982	.346	.578
Login IGLS per week	32.262	2	16.131	5.687	.041
Error	17.018	6	2.836		
Total	89168.000	10			
Corrected Total	54.400	9			

Dependent Variable: Experimental group post-test
a R Squared = .687 (Adjusted R Squared = .531) p<.05

The result is as follows: strongly agree (61.82%), agree (27.27%), neutral (10.91%), disagree (0%), and strongly disagree (0%), with a grand mean of 4.51. Over ninety percent students approve of the system’s attraction to their participation and intension shown as Table 7 through Table 9.

CONCLUSION

According to digital game based learning theory, the research used IGLS as an ancillary tool in

seventh graders’ history teaching. In the end of the experiment, the following three aspects were examined: the improvement in academic performance, increase in motivation, and positive learning attitude. First, IGLS benefits academic performance in junior high history learning. Second, IGLS especially provides low achievers with confidence and positive attitude. Third, IGLS arouses conspicuous improvement in both motivation and concentration. In conclusion, IGLS has positive effects on history learning, thus, it can be of assistance in history teaching.

Table 7. Satisfaction of IGLS’s interface

	Assessment Score					Mean	SD
	5	4	3	2	1		
1.The interface is easy to handle	63.64%	36.36%	0.00%	0.00%	0.00%	4.64	0.505
2.The content is clearly expressed	72.73%	27.27%	0.00%	0.00%	0.00%	4.73	0.467
3.The buttons are easy to click on	81.82%	18.18%	0.00%	0.00%	0.00%	4.82	0.405
4.Convenience of the design	63.64%	27.27%	9.09%	0.00%	0.00%	4.55	0.688

Table 8. Satisfaction of IGLS's

	Assessment Score					Mean	SD
	5	4	3	2	1		
1.The records do help me with learning	63.64%	27.27%	9.09%	0.00%	0.00%	4.55	0.688
2.The group competition help me with learning	63.64%	27.27%	9.09%	0.00%	0.00%	4.55	0.688
3.The online version help me with learning	54.55%	45.45%	0.00%	0.00%	0.00%	4.55	0.522
4.The message board makes communication easier for me	36.36%	36.36%	27.27%	0.00%	0.00%	4.09	0.831
5.I can handle the situation in group competition	63.64%	18.18%	18.18%	0.00%	0.00%	4.45	0.820

Table 9. Satisfaction of IGLS's motivation

	Assessment Score					Mean	SD
	5	4	3	2	1		
1.The game arouses my participation	63.64%	27.27%	9.09%	0.00%	0.00%	4.55	0.688
2.The game improves my performance	63.64%	36.36%	0.00%	0.00%	0.00%	4.64	0.505
3.The game eases my pressure in learning	72.73%	18.18%	9.09%	0.00%	0.00%	4.64	0.674
4. The game attracts my interest in history	54.55%	27.27%	18.18%	0.00%	0.00%	4.36	0.809
5.I think the system is suitable for learning history	54.55%	27.27%	18.18%	0.00%	0.00%	4.36	0.809

REFERENCES

Bruner, J. (1986). Play, thought and language. *Prospects: Quarterly Review of Education*, 16(1), 77–83. doi:10.1007/BF02197974

Bruner, J. S. (1960). *The process of education*. New York, NY: Vintage Books.

Chang, C. C. (2003). *Training & development fashion*. Paper presented at the 8th Network Operation and Management of the Community Learning Meeting.

Cheng, Y. M., & Chen, P. F. (2008, August 15-17). Building an online game-based learning system for elementary school. In *Proceedings of the International Conference on Intelligent Information Hiding and Multimedia Signal Processing*, Harbin, China.

Csikszentmihalyi, M. (1991). *Flow: The psychology of optimal experience*. New York, NY: Harper Perennial.

Green, B., Reod, J., & Bium, C. (1998). Teaching the Nintendo generation? Children, compute culture, and popular technologies. In Howard, S. (Ed.), *Wired UP* (pp. 19–42). London, UK: UCL Press.

Johnson, D. W., & Johnson, R. T. (1991). *Learning together and alone: Cooperative, competitive, and individualistic learning* (3rd ed.). Upper Saddle River, NJ: Prentice Hall.

Keller, J. M. (1983). Motivational design of instruction. In Reigeluth, C. M. (Ed.), *Instructional design theories and models: An overview of their current status* (pp. 384–434). Mahwah, NJ: Lawrence Erlbaum.

Lin, K.-C., Wu, T.-K., & Wang, Y.-B. (2011). Developing a web-based and competition-based quiz game environment to improve student motivation. *Journal of Network*, 6(5), 736-742.

Lippitt, R. (1981). Humanizing planned changes. In Meltzer, H., & Nord, W. (Eds.), *Making organizations humane and productive, a handbook for practitioners* (pp. 463-474). New York, NY: John Wiley & Sons.

Lu, Y.-C. (2000). *An empirical study on implementation of web-based community at university—Usages of learning community, factors and influences on learning performance*.

Mitchell, A., & Savill-Smith, C. (2004). *The use of computer and video games for learning: A review of the literature*. London, UK: Learning and Skills Development Agency.

Munn, N. L., Femald Dodge, L., & Femald, D. S. (1969). *Introduction to psychology*. Boston, MA: Houghton Mifflin.

Paras, B., & Bizzocchi, J. (2005). Game, motivation, and effect learning: An integrated model for educational game design. In *Proceedings of the Digital Games Research Association*.

Piaget, J. (1951). *Play, dreams and imitation in childhood*. New York, NY: W. W. Norton.

Prensky, M. (2001). *Digital game-based learning*. New York, NY: McGraw-Hill.

Prensky, M. (2003). Digital game-based learning. *Computers in Entertainment*, 1(1), 21. doi:10.1145/950566.950596

Reilly, W. J. (1983). *A manual for the use of the motivated strategies for learning questionnaire (MSLQ)*. Ann Arbor, MI: National Center for Research to Improve Postsecondary Teaching and Learning.

Taiwan Institute for Information Industry. (2010). *A survey on broadband usage in Taiwan*. Taipei, Taiwan: Author.

Taran, C. (2005). Motivation techniques in eLearning. In *Proceedings of the Fifth IEEE Interactional Conference on Advanced Learning Technologies* (pp. 617-619).

Vygotsky, L. S. (1966). Play and its role in the mental development of the child. In Cole, M. (Ed.), *Soviet developmental psychology* (pp. 76-99). New York, NY: M. E. Sharpe. doi:10.2753/RPO1061-040505036

Wang, C. (2003). Action research on establishing an online learning community via online peer-teaching. *Journal of Taiwan Normal University, Mathematics & Science Education*, 48(1), 119, 142.

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